

```
1) data(women)
```

```
women
```

```
x <- women$height
```

```
y <- women$weight
```

```
cor(x,y)
```

```
plot(x, y,
```

```
  main = "Height vs Weight",
```

```
  xlab = "Height",
```

```
  ylab = "Weight",
```

```
  pch = 19,
```

```
  frame = FALSE)
```

```
abline(lm(y~x), col="red")
```

```
data(mtcars)
```

```
mtcars
```

```
x <- mtcars$wt
```

```
y <- mtcars$mpg
```

```
cor(x,y)
```

```
plot(x, y,
```

```
  main = "Negative Correlation (wt vs mpg)",
```

```
  xlab = "Weight",
```

```
  ylab = "Mileage (mpg)",
```

```
  pch = 19,
```

```
  frame = FALSE)
```

```
abline(lm(y~x), col="blue")
```

```
2) LungCapData <- read.csv("G:/DA Lab Dataset/LungCap.csv", header = T)
```

```
LungCapData <- data.frame(LungCapData)
```

```
attach(LungCapData)
```

```
AgeGroups <- cut(
```

```
  LungCapData$Age,
```

```
  breaks = c(0, 13, 15, 17, 25),
```

```
  labels = c("<13", "14/15", "16/17", ">=18")
```

```
)
```

```
head(LungCapData)
```

```
boxplot(
```

```
  LungCapData$LungCap ~ LungCapData$Smoke,
```

```
  ylab = "Capacity",
```

```
  main = "Lung Capacity of Smokers Vs Non-Smokers",
```

```
  las = 1
```

```
)
```

```
boxplot(
```

```
  LungCapData$LungCap[LungCapData$Age >= 18] ~
```

```
  LungCapData$Smoke[LungCapData$Age >= 18],
```

```
  ylab = "Capacity",
```

```
  main = "Lung Capacity of Smokers Vs Non-Smokers (Age >= 18)",
```

```
  las = 1
```

```
)
```

```
boxplot(
```

```
  LungCapData$LungCap ~ LungCapData$Smoke * AgeGroups,
```

```
  ylab = "Capacity",
```

```
  xlab = "",
```

```
  main = "Lung Capacity of Smokers Vs Non-Smokers",
```

```
  col = c(4, 2),
```

```
  las = 2 )
```

3) airqual=airquality

summary(airquality)

airqual[!complete.cases(airqual),]

boxplot(airqual)

boxplot(airqual\$Ozone, horizontal = TRUE)

boxplot(airqual\$Wind, horizontal = TRUE)

Updated_airqual=subset(airqual,Ozone<130 & Wind<17)

boxplot(Updated_airqual)

airqual\$Ozone[is.na(airqual\$Ozone)]<-mean(Updated_airqual\$Ozone)

summary(airqual)

airqual\$Solar.R[is.na(airqual\$Solar.R)]<-mean(Updated_airqual\$Solar.R,na.rm = TRUE)

summary(airqual)

data_airquality=subset(airqual,Ozone<70 & Wind<17)

boxplot(data_airquality)

nrow(data_airquality)

data_airquality=subset(airqual,Ozone<70 & Wind<17 & Wind>2)

boxplot(data_airquality)

```
4) data("iris")
```

```
str(iris)
```

```
summary(iris)
```

```
set.seed(111)
```

```
ind <- sample(2, nrow(iris),  
             replace = TRUE,  
             prob = c(0.8, 0.2))
```

```
training <- iris[ind==1,]
```

```
testing <- iris[ind==2,]
```

```
install.packages("tmvnsim")
```

```
install.packages("psych")
```

```
install.packages("psych",dependencies = TRUE)
```

```
library(psych)
```

```
pairs.panels(training[,-5],  
             gap = 0,  
             bg = c("red", "yellow", "blue")[training$Species],  
             pch = 21)
```

```
pc <- prcomp(training[,-5],  
            center = TRUE,  
            scale. = TRUE )
```

```
attributes(pc)
```

```
pc$center
```

```
pc$scale
```

```
print(pc)
```

```
summary(pc)
```

```
5) rm(list = ls(all.names = TRUE))
```

```
x <- data.frame(  
  name = c("Bala", "Ganesh", "jeevan"),  
  age = c(43, 38, 42),  
  education = c(2.0, 4.2, 4.1)  
)
```

```
print(x)
```

```
age <- x$age
```

```
print(age)
```

```
edu <- x$education
```

```
print(edu)
```

```
M1 <- matrix(c(age, edu), nrow = 3, byrow = FALSE)
```

```
print(M1)
```

```
E1 <- dist(M1, method = "euclidean")
```

```
print(E1)
```

```
ageD <- age / 10
```

```
print(ageD)
```

```
M2 <- matrix(c(ageD, edu), nrow = 3, byrow = FALSE)
```

```
print(M2)
```

```
E2 <- dist(M2, method = "euclidean")
```

```
print(E2)
```

```
rangA <- max(age) - min(age)
```

```
mi_maA <- (age - min(age)) / rangA
```

```
print(mi_maA)
```

```
rangE <- max(edu) - min(edu)
```

```
mi_maE <- (edu - min(edu)) / rangE
```

```
print(mi_maE)
```

```
M3 <- matrix(c(mi_maA, mi_maE), nrow = 3, byrow = FALSE)
```

```
print(M3)
```

```
E3 <- dist(M3, method = "euclidean")
```

```
print(E3)
```

```
print("Data frame with different similarity measure matrix results")
```

```
print(x)
```

```
print(E1)
```

```
print(E2)
```

```
print(E3)
```

```
6) rm(list = ls(all.names = TRUE))

input = read.csv("G:/DA Lab Dataset/ghi.csv")

print(input)

x = cor(input[,c(2:7)])

print(x)

pairs(input[,c(2:7)])

y = x[6,1:5]

print(y)

x= sort(y,decreasing = TRUE)

print(x)

cnt =1

print("selected attributes")

while(cnt<=length(y))

{ if (y[cnt]>.5)

  print(y[cnt])

  cnt=cnt+1

}

7) data(iris)

str(iris)

install.packages("ClusterR")

install.packages("cluster")

library(ClusterR)

library(cluster)


iris_1 <- iris[,-5]

set.seed(240)

kmeans.re <- kmeans(iris_1, centers = 3, nstart = 20)

kmeans.re
```

```
kmeans.re$cluster
```

```
cm <- table(iris$Species, kmeans.re$cluster)
```

```
cm
```

```
plot(iris_1[c("Sepal.Length", "Sepal.Width")])
```

```
plot(iris_1[c("Sepal.Length", "Sepal.Width")],
```

```
col = kmeans.re$cluster)
```

```
plot(iris_1[c("Sepal.Length", "Sepal.Width")],
```

```
col = kmeans.re$cluster,
```

```
main = "K-means with 3 clusters")
```

```
kmeans.re$centers
```

```
kmeans.re$centers[, c("Sepal.Length", "Sepal.Width")]
```

```
points(kmeans.re$centers[, c("Sepal.Length", "Sepal.Width")],
```

```
col = 1:3, pch = 8, cex = 3)
```

```
y_kmeans <- kmeans.re$cluster
```

```
clusplot(iris_1[, c("Sepal.Length", "Sepal.Width")],
```

```
y_kmeans,
```

```
lines = 0,
```

```
shade = TRUE,
```

```
color = TRUE,
```

```
labels = 2,
```

```
plotchar = FALSE,
```

```
span = TRUE,
```

```
main = paste("Cluster iris"),
```

```
xlab = 'Sepal.Length',
```

```
ylab = 'Sepal.Width')
```



```
8) install.packages("arules")
```

```
install.packages("arulesViz")
```

```
installed.packages("RColorBrewer")
```

```
library(arules)
```

```
library(arulesViz)
```

```
library(RColorBrewer)
```

```
SupMarTra<-read.transactions("G:/DA Lab Dataset/Snakyy.csv",format="basket",sep=",")
```

```
summary(SupMarTra)
```

```
dim(SupMarTra)
```

```
View(SupMarTra)
```

```
rules=apriori(SupMarTra)
```

```
inspect(rules)
```

```
rules=apriori(SupMarTra, parameter = list(supp=.3,conf=.8))
```

```
inspect(rules)
```

```
rules=apriori(SupMarTra,parameter = list(supp=.5,conf=.8))
```

```
inspect(rules)
```

```
plot(rules)
```

```
plot(rules,method = "graph")
```

```
9) require("class")
```

```
require("datasets")
```

```
data("iris")
```

```
str(iris)
```

```
summary(iris)
```

```
head(iris)
```

```
set.seed(99)
```

```
rnum <- sample(rep(1:150))
```

```
rnum
```

```
iris <- iris[rnum,]
```

```
head(iris)
```

```
normalize <- function(x) { return((x - min(x)) / (max(x) - min(x))) }
```

```
iris.new <- as.data.frame(lapply(iris[, c(1,2,3,4)], normalize))
```

```
head(iris.new)
```

```
iris.train <- iris.new[1:130,]
```

```
iris.train.target <- iris[1:130,5]
```

```
iris.test <- iris.new[131:150,]
```

```
iris.test.target <- iris[131:150, 5]
```

```
model1 <- knn( train = iris.train, test = iris.test, cl = iris.train.target, k = 16)
```

```
model1
```

```
table(iris.test.target,model1)
```

```
10) df <- data.frame(hours = c(1,2,4,5,5,6,6,7,8,10,11,11,12,12,14),  
  score = c(64,66,76,73,74,81,83,82,80,88,84,82,91,93,89))
```

```
head(df)
```

```
plot(df$hours,df$score,main = 'Hours studied vs Exam score',xlab = 'Hours Studied', ylab =  
'Exam score')
```

```
boxplot(df$score,main = 'Boxplot of Exam scores',ylab = 'Exam score')
```

```
model <- lm(score~hours,data=df)
```

```
summary(model)
```

```
res <- resid(model)
```

```
plot(fitted(model),res,main = 'Residuals vs Fitted',xlab = 'Fitted values',ylab = 'Residuals')
```

```
abline(h=0,col='red')
```

```
qqnorm(res,main = 'Normal Q-Q Plot')
```

```
qqline(res,col='red')
```